

The use of surfactants in de-inking paper for paper recycling

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Many de-inking surfactants have been claimed in patents. Alcohol ethoxylates and alkylphenol ethoxylates are commonly used in wash de-inking whereas alcohol alkoxyates and fatty acid alkoxyates are commonly used in flotation de-inking and combined flotation/wash de-inking processes. Dispersed ink particle redeposition onto cellulose fibers is a significant problem. Recent developments have occurred in the improvement of de-inking chemicals and in the understanding of surface chemistry.

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Abbreviations

EO ethylene oxide
PO propylene oxide

Introduction

Efficient ink removal is necessary to recycle used paper into high value products for newspapers, magazines, and other printing applications. Surface chemistry plays an important role in some de-inking unit operations. Major unit operations involved in de-inking recovered paper are summarized in Table 1. Paper de-inking consists of two processes: ink detachment from cellulose fibers and separation of dispersed ink particles from the pulp slurry.

Paper mills use surfactants to promote both of these processes [1]. Mechanisms of ink detachment from cellulose fibers are summarized in Table 2.

Like soils encountered in laundering, inks differ in composition and surface chemistry [2]. As a result, they respond differently to the mechanical forces and added chemicals used in pulping, washing, and flotation. Mechanical forces are more important than in laundering because paper must be disaggregated into dispersed, individual cellulose fibers (pulp). This disaggregation aids in ink detachment. Mechanical energy is also required to disintegrate toner inks into particle sizes that can be removed by washing and flotation.

Performance criteria for de-inking surfactants include the following: effectiveness of ink detachment from cellulose fiber during pulping; effectiveness in promoting the separation of dispersed ink particles from cellulose fibers in various unit operations; wash de-inking—minimal foaming; flotation de-inking—controlled foaming sufficient to promote good collection of ink in the froth layer

but not enough to cause excessive fiber loss or foaming problems elsewhere in the mill; and minimal interference with papermaking and properties of the recycled paper. In addition, the de-inking chemicals should not interfere with process water clarification.

The most common measurements of de-inking process performance are paper brightness and the number of visible ink specks on a sheet surface. De-inked paper sheet brightness is measured using a reflectance device called a brightness meter. Computer analysis of video images of paper surfaces can determine the number of ink specks, their size, and size distribution.

The focus of this review will be on types of surfactants currently used in commercial de-inking mills, the effect of surfactant chemical structure and properties on de-inking effectiveness, and the role of surface chemical phenomena in de-inking.

Pulping

The roles of the various chemicals used in pulping are summarized in Table 3. Disaggregation of paper into individual cellulose fibers and ink detachment from fibers occurs in the pulper. For most inks, mechanisms of ink detachment are thought to be similar to those proposed for liquid soil detachment from fabrics [3,4]. Toner and ultraviolet-cured inks are thought to behave similarly to solid soils in laundering [5].

Typical pulping process temperatures for de-inking newsprint are 40–60°C. Mills often process old office papers at 50–90°C.

Dispersed ink particles formed during pulping must be removed to prevent their redeposition onto the cellulose. Two unit operations that do so by relying on surface chemistry are wash de-inking and flotation de-inking.

Wash de-inking

In wash de-inking, much of the water containing dispersed ink is drained off the pulp. The role of the surfactant is largely to disperse ink to the proper size range in the pulper for later removal by washing. Wash de-inking is most effective over a particle diameter range of 3–25 microns [6]. For hydrophobic particles such as carbon black and other pigments, longer hydrophobe, high HLB surfactants are most effective [7]. The most commonly used wash de-inking surfactants are alcohol ethoxylates and alkylphenol ethoxylates [8]. Washing is used with highly dispersible inks.

Surfactant properties and wash de-inking

Nonionic surfactant cloud point is a critical property in determining surfactant detergency at various clothing wash

Table 1**Major paper de-inking mill unit operations.**

Pulping disaggregates paper into individual cellulose fibers.

Screening is used to remove relatively large particles including staples, paper clips, miscellaneous debris, larger toner ink particles, and large adhesive particles. Screens with slot widths of 152 microns are often used in mills. Efficient screening depends on differences in size and geometry between the long, thin cellulose fibers and the inks and other contaminants.

Mechanical screening with cyclone devices is used to remove inks and adhesive particles. Separation is based largely on density differences with devices designed to remove objects that are either more dense (toner inks) or less dense (adhesive particles) than water.

Flotation removes somewhat larger ink particles and is most effective when the ink is considerably more hydrophobic than the cellulose fibers. Surfactants added to the pulper or to the flotation cell are used to promote the formation of ink particles in the size range over which flotation is most effective and to render ink-particle surfaces more hydrophobic.

Washing involves the drainage of water from the suspended cellulose fibers. Relatively small ink particles dispersed by surfactant in the aqueous medium are thus separated from cellulose. Surfactants are added to the pulper to promote the formation of small, relatively hydrophilic dispersed ink particles.

Table 2**Ink detachment mechanisms*.**

Surfactant-promoted solubilization into the aqueous pulping medium.

Surfactant-promoted wettability alteration of cellulose surfaces promoting ink detachment and emulsification.

Cellulose fiber swelling. This reduces ink adhesion to the fiber. Fiber swelling is promoted by high pH.

Cellulose fiber bending and interfiber abrasion promoted by mechanical agitation.

*These processes occur in the pulper. The last process also occurs in recently developed units called kneaders and dispergers.

temperatures. Removal of nonpolar soils from cotton cloth is greatest at a wash temperature approximately 15–25°C above the surfactant cloud point [3]. The soil used in these studies was mineral oil containing dispersed carbon black particles — essentially the composition of letterpress ink. In contrast, maximum removal of polar oil phases from cotton fabrics occurs when the wash temperature is 0–10°C below the surfactant cloud point [9].

Maximum letterpress newsprint de-inked pulp brightness occurred when the process temperature was within 5°C of the de-inking surfactant cloud point [8]. Optimum washing temperature is a function of ink composition. Maximum wash de-inking effectiveness for a more polar ink on books occurred when the process temperature was 20–30°C below the de-inking surfactant cloud point [10]. Thus, this ink behaved similarly to a polar soil in laundering.

Optimum surfactant HLB for wash de-inking of letterpress printed newsprint is 14.5–15.5 [7]. Optimum HLB for de-inking ledger paper, a more polar ink than letterpress ink, was 11.5–12 whereas that for toner inks was 10–11 [10]. Newsprint wash de-inking efficiency was correlated with surfactant solution interfacial tension against a model ink [11].

Flotation de-inking

The efficiency of ink-air bubble attachment is, in part, a function of surfactant chemistry (see below). The ink particle diameter range over which flotation de-inking is effective is somewhat larger than for wash de-inking, 20–200 microns in diameter, with greatest effectiveness in the 30–80 micron size range [6,12–14]. The bubbles rise to the top of the liquid and form a foam layer at the surface where they are removed by various mechanical means.

Surfactants need to provide a controlled amount of foaming during flotation that is sufficient to form a froth layer stable enough to trap ink particles and with long enough a lifetime for the froth to be separated from the pulp slurry, but not so stable that excessive amounts of foam cause mill operating problems and excessive fiber loss during flotation.

Three factors must be satisfied for efficient flotation de-inking: the ink particles removed must be of the proper size range and surface chemistry (hydrophobic); air bubbles must be of the proper size range and be introduced into the flotation cell in adequate numbers; there must be sufficient flotation cell turbulence to allow good mixing of air and pulp without promoting too rapid a rate of air bubble collapse.

Table 3

Chemicals used in pulping.

Chemical	Purpose	Concentration (relative to dry paper weight)	
		newsprint, newsprint/magazine	office paper
Surfactant	Disperses ink particles to the proper size range	0.1–1.0%	0.1–1.5%
Fatty acid	Seldom added to pulper, disperses ink particles to the proper size range	0–1.5%	0–1.5%
NaOH*	Promotes swelling and dispersion of cellulose fibres, saponifies ink binder resins	0.25–1.0%	1.0–1.5%
Na silicate	Dispersant, raises pH to swell and disperse cellulose particles, helps stabilize H ₂ O ₂	0.25–1.5%	–
H ₂ O ₂	Retards yellowing of lignin-containing pulp	0.5–1.5%	–
Chelants	Stabilize H ₂ O ₂	0–0.5%	–

*Sodium hydroxide may also promote peptization. This occurs when ester-based ink binder resins are hydrolyzed and ink particles break apart. Peptization may also play a role in the behavior of toner inks containing styrene-acrylate copolymer resins.

The second and third requirements are necessary for a sufficient number of ink-air bubble collisions to occur to allow efficient ink removal. Ink removal during flotation follows modified first order kinetics [12–14].

A minimum air bubble diameter of 0.3 mm is needed for the air bubble to rise through the pulp slurry [15]. Data from mineral ore flotation suggest that the optimum air bubble size is about five times that of the ink particles being removed [16,17]. As pulping produces a range of ink particle sizes, flotation cell air injectors need to produce a range of bubble sizes to remove ink efficiently. High flotation cell turbulence is necessary to remove sub-visible (<ca. 70 micron in diameter) ink particles efficiently. No increase in ink removal efficiency was observed for larger ink particles at high flotation cell turbulence. Excessive cell turbulence, however, can detach large ink particles from air bubbles [18].

Ink particles that are too small, such as flexographic ink particles (many of which are less than one micron in diameter), cannot adhere to air bubble surfaces due to interfacial tension and electrical double layer forces [16,17]. Reducing the flotation pH and thus the degree of ionization of flexographic ink binder resins can increase the efficiency of flotation de-inking [19,20] presumably by reducing ink particle dispersion. A solvent-encapsulated surfactant promotes agglomeration of ink particles less than 5 microns in size [21], thus increasing flotation ink removal efficiency.

Flotation ink removal efficiency for newsprint increases with decreasing process water/ink interfacial tension and

decreasing negative value of the zeta-potential imparted by the de-inking surfactant to the ink particle surface [22]. Contact angle studies indicate more effective newsprint ink wetting by surfactant solutions is correlated with increased flotation ink removal efficiency [23]. Similar tests indicated the most effective toner wetting surfactant was the most effective de-inking agent in an office paper wash-flotation deinking mill [24].

Calcium soaps of fatty acids commonly are used in flotation de-inking [1*]. They promote the formation of relatively large, hydrophobic ink particles of the correct size for efficient removal by flotation. Chemical addition is usually immediately prior to the flotation operation. Fatty acids are commonly used in Europe and Canada. Their use is less common, however, in the United States and Japan because washing steps are commonly used in addition to flotation. Calcium salts of fatty acids size ink particles too large for efficient removal by wash de-inking. Usually, the process water is not hard enough to completely convert fatty acids to their calcium salts and calcium chloride must be added. Calcium ion levels in the aqueous phase should be above 100–150 ppm.

The calcium ion is thought to be simultaneously bonded to the fatty acid and to a negatively charged ink particle [25]. (Ink binder resins such as acrylate polymers give rise to these negative charges). The fatty acid renders the ink particle hydrophobic, thus promoting its attachment to air bubbles. This mechanism, however, is under some dispute. What does seem clear is the important role of calcium ions. If insufficient calcium ions are present, the sodium salt of the fatty acid will emulsify the ink

into small particles poorly sized for removal by flotation. Excessive water hardness reduces flotation ink removal effectiveness [26].

Optimum fatty acid carbon chain length for newsprint flotation de-inking appears to be sixteen [27]. Saturated fatty acid is reported to be more effective than unsaturated fatty acids [28].

During pulping, water-based flexographic inks tend to form dispersed ink particles too small for efficient ink removal by flotation. A low pH, however, or a high concentration of calcium ions at alkaline pH can promote ink aggregation or retard dispersion thus improving flotation de-inking efficiency [29]. Inclusion of a fatty acid calcium soap further improved flotation ink removal at alkaline pH. This was attributed to hydrophobization of ink particle aggregates by calcium soap encapsulation [29]. It should be noted that these experiments were performed in the absence of cellulose fibers.

Advancing and receding contact angle studies indicate that the hydrophobicity of toner surfaces is decreased after aqueous wetting [30,31]. Increasing pulper pH also decreased toner hydrophobicity [30]. Froth stability increases and toner particle hydrophobicity decreases with increasing surfactant concentration [32]. The first effect favors higher ink removal efficiency whereas the second effect operates in the opposite direction.

Models of flotation

On a microscopic level, the flotation process occurs in a series of steps: first, the sufficiently close approach of a bubble to a suspended ink particle for collision to occur; second, bubble-particle attachment resulting from rupture of the thin liquid film separating the ink particle and the bubble (dewetting); third, transport of the air-bubble-ink-particle aggregate to the froth layer followed by froth removal from the pulp slurry. For this to occur, the ink-particle-bubble aggregate must have sufficient stability that the ink particle remains attached to the air bubble and the air bubble does not break before becoming part of the froth.

Elegant theories of the forces involved in flotation have been developed. A hydrodynamic model for the approach and collision of an air bubble and ink particle resulting in ink attachment to the bubble considers long- and short-range hydrodynamic forces and short-range nonhydrodynamic interactions [33,34].

Fluid flow creates long-range hydrodynamic interactions of bubbles and ink particles. Short-range hydrodynamic forces include squeezing flow as fluid in the gap between the ink particle and air bubble is expelled, shear forces or torque of the air bubble and ink particle on each other, and torque created by possible rotation of the air bubble and ink particle [35,36]. Nonhydrodynamic interactions

(including surface chemistry considerations) are combined in a single parameter called the critical gap [33,34]. This is the maximum distance at which a strong attractive force acts (essentially instantaneously) resulting in ink particle attachment to the air bubble. The individual forces determining the critical gap include surface tension, London-van der Waals dispersion forces and electrostatic effects.

A second model [37••] considers these forces explicitly in analyzing rupture of the liquid film separating the ink particle and an approaching air bubble [38]. This disjoining film rupture model is designed to determine the circumstances in which the thin film separating an ink particle and an approaching air bubble becomes unstable and ruptures leading to ink-particle-bubble contact and attachment. The model permits study of the forces influencing film rupture, the effect of surfactants, and the time of film thinning and rupture. The Maragoni effect increases film rupture time [39].

The concept of hydrophobic attractive forces has been discussed recently [40,41] and is incorporated into the model. This inclusion is consistent with the observation that maximum flotation ink removal efficiency for a homologous series of surfactants shifts to lower surfactant concentration with increasing linear hydrophobe carbon number, that is, increasing hydrophobicity [32]. Toner particle zeta-potential values and advancing and receding measurements indicate surfactant adsorption onto toner particles occurs via the hydrophobe rendering toner particles less hydrophilic [32] (hydrophobic attraction forces may also play a role in the adsorption of toner particles and oil-based inks by plastics prior to their removal by screens and mechanical cleaners [42,43]—agglomeration of toner particles by hydrophobic liquid bridging [44] also appears to involve hydrophobic attraction forces).

Atomic force microscopy has been used to measure the attractive forces between air bubbles and hydrophobized spherical glass beads and less regularly shaped toner particles [45,46•]. A bubble is held on the tip of a syringe needle. The toner particle is glued to the tip of an atomic force microscope cantilever arm. The force of attraction between the bubble and a particle is measured as the syringe with its attached bubble is moved toward the particle (using a piezoelectric device to control bubble movement). After toner particle-air bubble attachment has taken place, the force required for toner particle detachment from the air bubble can be measured as the syringe with its attached air bubble is moved away from the toner particle.

Together, the hydrodynamic and disjoining film models can predict the effect of surface chemical forces on the critical gap. These two models may be combined in a statistical or global flotation model to calculate flotation ink removal efficiency [47–49].

Comparison to laboratory flotation results indicated that two capture radius values are needed for each ink particle size to obtain excellent agreement between global model predictions of flotation efficiency and observed results in laboratory and pilot plant flotation experiments. Thus, it appears two different types of particles are formed in each particle size range. Microscopy studies indicated that the cause of this difference was incomplete defibering of toner ink particles on pulping [46*,48–50,51*]. In any given particle size range, toner particles having no attached fibers exhibited one capture radius value whereas toner particles containing attached cellulose fibers exhibited another. The lower flotation efficiencies observed for toner particles containing attached cellulose fibers may be due to either the more hydrophilic character of these particles compared to fiber-free toner particles or to steric interference of toner attachment to air bubbles by the attached cellulose fibers.

Another theoretical model considers buoyancy, gravity, capillary forces, and external turbulence on the stability of ink-air-bubble aggregates [52]. The model indicates important factors in determining aggregate stability are: liquid surface tension, ink particle size and surface free energy, air bubble size, and flotation cell turbulence. The focus of a population balance model is the effect of ink particle size and ink-bubble aggregate stability on flotation efficiency as a function of time [53]. The emphasis of another flotation modeling study was the effect of flotation cell turbulence on toner ink removal rate [44].

Combined flotation and wash de-inking

US de-inking mills use both wash and flotation units in series. Usually, the flotation unit is the first in the series. Surfactants have been developed to work well in both washing and flotation. These are generally proprietary reaction products of fatty alcohols or fatty acids and both ethylene oxide (EO) and propylene oxide (PO) [1*].

Patents suggest the optimum ratio of EO and PO is 2:1 by weight, whereas the optimum hydrophobe carbon chain length is 16–18 [P1,P2]. EO:PO ratios ranging from 1:2 to 4:1, however, have been claimed [P3,P4]. Generally, the EO and PO are reacted simultaneously with the substrate, resulting in a distribution of ethoxy and propoxy groups related to their relative reactivities and the relative amounts of each used in the synthesis. The separate addition, however, of EO and PO to form blocks of these units in the final surfactant has also been reported [54,55]. Random addition of EO and PO to a fatty alcohol produced a surfactant providing higher flotation newsprint ink removal efficiency, lower interfacial tension with offset newsprint ink, a more negative ink particle zeta-potential, and higher foaming than did block addition of the same amount of EO and PO [55].

Linear hydrophobes are preferred [54,55,P1–P4] although twin-tailed hydrophobe alkoxylates have also been claimed

[56*]. Copolymers of EO and PO have been used in flotation de-inking [P5].

Other concerns

The de-inked pulp is now ready to be converted to paper. Surfactant adsorption onto cellulose fibers and its effects on papermaking and final paper properties is a major concern [57,58]. De-inking surfactants affect process water clarification [59*,60] and process water recycling. Incomplete ink removal on clarification can lead to ink redeposition later when using recycled process water [61]. The use of laundering anti-redeposition agents [62] and ultrafiltration techniques [63] has been evaluated to reduce this problem.

Conclusions

Surfactants play a key role in paper de-inking, promoting ink detachment from cellulose fibers and separation of dispersed ink particles from pulped wastepaper. Active research is in progress to develop improved de-inking chemicals. A better understanding of surface chemistry is needed to support this development. This understanding, particularly in flotation de-inking, is rapidly being obtained.

References and recommended reading

Papers of particular interest, published within the annual period of review, have been highlighted as:

- of special interest
- of outstanding interest

1. Borchardt JK: **Recycling (paper)**. In *Kirk-Othmer Encyclopedia of Chemical Technology*, vol 21. New York: John Wiley & Sons Inc; 1997:11–22.
- The engineering of the unit operations and the chemicals used in each operation are reviewed. De-inking surfactant technology is briefly reviewed. Economic aspects and the future of paper recycling are discussed.
2. Borchardt JK: **Ink types: the role of ink in deinking**. *Prog Paper Recycling* 1995, 5:81–90.
3. Kissa E: **Kinetics and mechanisms of soiling and detergency**. In *Detergency: Theory and Technology*. Edited by Cutler WG, Kissa E. New York: Marcel Dekker Inc; 1987:193–331.
4. Borchardt JK: **Mechanistic insights into deinking**. *Colloid Surf A* 1994, 88:13–25.
5. Morsink JBW, Daane GFR: **Surfactant-aided detachment of ink from impact- and non-impact printed paper**. In *Proc TAPPI Pulping Conference*. Atlanta: TAPPI Press; 1992:963–975.
6. Shrinath A, Szewczak JT, Bowen JJ: **A review of ink-removal techniques in current deinking technology**. *TAPPI J* 1991, 74:85–93.
7. Turai LL, Williams LD: **Effect of HLB factor of nonionic surfactants on deinking efficiency**. *TAPPI J* 1977, 60:167–168.
8. Wood DL: **Ethoxylates and other nonionics as surfactants in the deinking of waste paper**. In *Proc TAPPI Pulping Conference*. Atlanta: TAPPI Press; 1982:435–446.
9. Raney KH, Benson HL: **The effect of polar soil components on the phase inversion temperature and optimum detergency conditions**. *J Am Oil Chem Soc* 1990, 67:722–729.
10. Borchardt JK, Rask JH: **Macro- and microscopic deinking studies of electrostatic ink containing furnishes**. In *Proc TAPPI Recycling Symposium*. Atlanta: TAPPI Press; 1993:839–873.
11. Borchardt JK, Tutunjian PN, Rask JH, Prieto NE: **Novel methods for laboratory evaluation of deinking surfactants**. *TAPPI*

- Contaminants and Problems in Paper Recycling Seminar. Cincinnati, USA; 28-30 April, 1992. Atlanta: TAPPI Press; 1992:85-112.*
12. Vidotti RM, Johnson DA, Thompson EV: Comparison of bench scale and pilot plant flotation of photocopied office waste paper. *Proc TAPPI Pulping Conference. Atlanta: TAPPI Press; 1992:643-652.*
 13. Vidotti RM, Johnson DA, Thompson EV: Repulping and flotation studies of laser-printed office waste paper. Part I. Flotation. *Prog Paper Recycling* 1993, 3:39-49.
 14. Julien Saint Amand F, Perrin B: The effect of particle size on ink and speck removal efficiency. *Pulp Paper Can* 1993, 10:25-28.
 15. Linck E, Britz H: Ink and speck removal efficiency - a matter of the right flotation cells. *Proc TAPPI Pulping Conference. Atlanta: TAPPI Press; 1990:123-131.*
 16. Li D, Fitzpatrick JA, Slattery JC: Rate of collection of particles by flotation. *Ind Eng Chem Res* 1990, 29:955-967.
 17. Ferguson LD: A review of flotation deinking technology. *Prog Paper Recycling* 1991, 1:17-23.
 18. Julien Saint Amand F: Hydrodynamics of flotation: experimental studies and theoretical analysis. *Proc TAPPI Recycling Symposium. Atlanta: TAPPI Press; 1997:219-241.*
 19. Galland G, Vernac Y: De-inking of wastepaper containing water-based flexo-printed newsprint. *Pulp Paper Can* 1993, 94:71-75.
 20. Jarrehult B, Lindqvist M, Hanecker D, Phan-Tri D: Chemical influences on the deinkability of flexo-printed waste paper. *Wochenbl Papierfabr* 1991, 119:811-818.
 21. Sain MM, Marchildon L, Daneault C: A correlation between ink separation chemistry and deinked paper properties affected by micro-ink flotation. *Prog Paper Recycling* 1995, 5:54-64.
 22. Masamizu K, Tai Y, Hagiwara M, Ukigai T: Development of deinking agents for flotation systems. *Proc TAPPI Recycling Symposium. Atlanta: TAPPI Press; 1994:39-52.*
 23. Okada E, Ishibashi Y, Takahishi H: Recent developments in Japanese deinking and deinking agent. *Proc 2nd Research Forum on Recycling. Canadian Pulp and Paper Association. Quebec, Canada; 5-7 October, 1993. Montréal, Canada: Canadian Pulp and Paper Association; 1993:31-36.*
 24. Borchardt JK, Lott VG: Deinking toner containing furnishes. Part 3. Are microscopic ink particles formed on pulping? *Proc TAPPI Recycling Symposium. Atlanta: TAPPI Press; 1995:17-36.*
 25. Hornfeck K: Chemie und Wirkungsweise der Tenside als Sammlerchemikalie für Druckfarben und Füllstoffe im Flotations-Deinking-Prozess. *Wochenblatt für Papierfabrikation Bd* 1985, 17:646-649. [Title translation: Chemistry and action of surfactants as collecting agents for printing inks and fillers in the flotation process.]
 26. Baumgarten HL, Grossmann H: Research activities in paper recycling at the papertechnische stiftung. Munich, Germany. *Proc TAPPI Recycling Symposium. Atlanta: TAPPI Press; 1994:181-186.*
 27. Mak N, Stevens J: Characteristics of fatty acid as an effective flotation deinking collector. *Proc 2nd Research Forum on Recycling. Canadian Pulp and Paper Association. Quebec, Canada, 5-7 October 1993. Montréal, Canada: Canadian Pulp and Paper Association; 1993:145-152.*
 28. Marchildon L, Bonnelly B, Lapointe M: The effect of double bonds present in the surfactant on the deinking efficiency of xerographic paper. *J Pulp Paper Sci* 1993, 19:156-159.
 29. Doris GM, Nguyen N: Flotation of model inks. Part II. Flexo ink dispersions without fibres. *Proc 2nd Research Forum on Recycling. Canadian Pulp and Paper Association. Quebec, Canada; 5-7 October 1993. Montréal, Canada: Canadian Pulp and Paper Association; 1993:13-22.*
 30. Ling TF, Richman SK: A study of correlation between surface properties and behavior of toner inks in the pulper. *Proc TAPPI Recycling Symposium. Atlanta: TAPPI Press; 1996:349-362.*
 31. Drellich J, Azevedo MAD, Miller JD, Dryden P: Hydrophobicity and elemental composition of laser-printed toner films. *Prog Paper Recycling* 1996, 5:31-38.
 32. Eppele M, Schmidt DC, Berg JC: The effect of froth stability and wettability on the flotation of a xerographic toner. *Colloid Polym Sci* 1994, 272:1264-1272.
 33. Pan R, Bousfield DW, Thompson EV: Modeling particle/bubble dynamics and adhesion in air bubble/solid particle/liquid systems. *Proc TAPPI Pulping Conference. Atlanta: TAPPI Press; 1992:941-956.*
 34. Paulsen FG, Pan R, Bousfield DW, Thompson EV: The dynamics of bubble/particle approach and attachment during flotation and the influence of short-range nonhydrodynamic forces on disjoining film rupture. *Proc 2nd Research Forum on Recycling Canadian Pulp and Paper Association. Montréal: Canadian Pulp and Paper Association; 1993:1-12.*
 35. Jeffrey J, Ohnishi Y: Calculation of the resistance and mobility functions for two unequal rigid spheres in low-reynolds-number flow. *J Fluid Mech* 1984, 139:261-290.
 36. Kim S, Karila SJ: *Microhydrodynamics: Principles and Selected Applications*. Stoneham, USA: Butterworth-Heinemann; 1991.
 37. Paulsen FG, Pan R, Bousfield DW, Thompson EV: The dynamics of bubble/particle attachment during flotation and the application of two disjoining film rupture models. part i. nondraining model. *J Colloid Interface Sci* 1996, 178:400-410.
 - A nondraining model for bubble film thinning is developed using a sinusoidal film profile. Equations are developed for three boundary conditions: free bubble surface, tangentially immobile bubble surface, and nonconstant surface tension. Besides surface tension, forces considered include viscous, London-van der Waals, dispersion, and hydrophobic attraction forces. If one associates the characteristic wavelength with ink particle dimensions, the model predicts particles of intermediate size are preferentially floated. This is consistent with many observed results reported by different research groups.
 38. Hwang C, Chang C, Chen J: On the rupture process of thin liquid films. *J Colloid Interface Sci* 1993, 159:184-188.
 39. Sharma A, Ruckenstein E: An analytical nonlinear theory of thin film rupture and its application to wetting films. *J Colloid Interface Sci* 1986, 113:456-479.
 40. Schmoller BK, Luttrell GH, Yoon R-H: Modelling of bubble - particle aggregation. In *Dispersion and Aggregation Fundamentals and Applications*. New York: Engineering Foundation; 1994:537-550.
 41. Yoon R-H, Ravishanker SA: An application of extended DLVO theory, effect of octanol on the long-range hydrophobic forces between dodecylamine - coated mica surfaces. *J Colloid Interface Sci* 1994, 166:215-224.
 42. Liu J, Muvundamina M: Role of polymers in de-inking mixed paper. *Proc TAPPI Pulping Conference. Atlanta: TAPPI Press; 1996:157-170.*
 43. Chu Y, Muvundamina M, Klungness J: Deinking of business bulk mail and mixed papers by use of polymer scavengers. *Prog Paper Recycling* 1995, 5:17-29.
 44. Snyder BA, Berg JC: Liquid bridge agglomeration: a fundamental approach to toner deinking. *Proc TAPPI Recycling Symposium. Atlanta: TAPPI Press; 1994:277-281.*
 45. Paulsen FG, Berg SR, Vidotti RM, Johnson DA, Thompson EV: Measurement of long-range hydrophobic attraction forces and their relationship to deinking flotation. Part I. *Proc TAPPI Pulping Conference. Atlanta: TAPPI Press; 1996:145-156.*
 46. Paulsen FG, Berg SR, Vidotti RM, Johnson DA, Thompson EV: • Measurement of long-range hydrophobic attraction forces and their relationship to deinking flotation. Part II. *Proc TAPPI Recycling Symposium. Atlanta: TAPPI Press; 1997:41-52.*
 - Atomic force microscopy is used to demonstrate that hydrophobic attraction plays a role in air bubble-toner interactions and bubble-toner attachment during flotation de-inking. The atomic force microscopy apparatus developed for this study is described in detail. Air bubble-toner attachment usually occurs at sharp projections of the toner particles, not flat surfaces.
 47. Pan R, Paulsen FG, Johnson DA, Bousfield DW, Thompson EV: A global model for predicting flotation efficiencies: model results and experimental studies. *Proc TAPPI Pulping Conference. Atlanta: TAPPI Press; 1993:1155-1164.*
 48. Pan R, Paulsen FG, Johnson DA, Bousfield DW, Thompson EV: A global model for predicting flotation efficiency. Part I: Model results and experimental studies. *Tappi J* 1996, 79:177-185.
 49. Pan R, Paulsen FG, Kocer H, Nerez R, Johnson DA, Bousfield DW, Thompson EV: A global model for predicting flotation efficiencies. Part II: particle size and flotation rate predictions, and experimental studies and comparisons. *Proc TAPPI Recycling Symposium. Atlanta: TAPPI Press; 1994:291-302.*

50. Johnson DA, Thompson EV: **Fiber/toner detachment during repulping of mixed office waste containing photocopied and laser printed paper.** *TAPPI J* 1995, 80:41-46.
51. Berg SR, Johnson DA, Thompson EV: **Toner detachment during repulping of laser printed office copy paper.** *Tappi J* 1997, 80:171-179.
Toner ink particle dispersion increases with increasing wastepaper pulping consistency and lower process temperature. Fiber detachment from toner is little influenced by operating conditions (including process temperature).
52. Hou MJ, Hui SH: **Interfacial phenomena in deinking. I. Stability of ink particle - air bubble aggregates in flotation deinking.** *Proc TAPPI Pulping Conference*. Atlanta: TAPPI Press; 1993:1125-1142.
53. Heindel TJ, Bloom F: **New measures for maximizing ink particle removal in a flotation cell.** *Proc TAPPI Recycling Symposium*. Atlanta: TAPPI Press; 1997:101-113.
54. Masamizu K, Egawa J, Hagiwara M, Kawai K: **Development of deinking agents for flotation systems.** *Proc TAPPI Recycling Symposium*. Atlanta: TAPPI Press; 1997:435-451.
55. Masamizu K, Egawa J, Hagiwara M, Kawai K: **Development of deinking agents for flotation systems: Part 2.** *Proc TAPPI Recycling Symposium*. Atlanta: TAPPI Press; 1995:317-326.
56. Masamizu K, Egawa J, Hagiwara M, Ukigai T: **Development of deinking agents for flotation systems. part 3.** *Proc TAPPI Recycling Symposium*. Atlanta: TAPPI Press; 1996:229-237.
The flotation de-inking effectiveness of fatty acids, fatty acid alkoxylates, and fatty alcohol alkoxylates was compared. A fatty alcohol alkoxylate was the most effective in removing offset ink from pulp by flotation. Flotation ink removal efficiency increased with decreasing de-inking surfactant solution-ink contact angle.
57. Fernández C, Garnier G: **Retention of fatty acid soaps during recycling. Part I: A study using packed beds of pulp fibres.** *J Pulp Paper Sci* 1997, 23:J144-J152.
58. Dorris GM, Dagenais M, Douek M, Boegh K, Allen J: **Use of distilled tall oil as a flotation collector in deinking.** *Proc 83rd Annual Meeting, Technical Section, CPPA*. Montréal: Canadian Pulp and Paper Association; 1997:B14-B22.

59. Guerro GJ, Schroeck JJ, Hsu NN-C, Errigo L: **Effect of Deinking Agents on Clarification Chemicals.** *Proc TAPPI Recycling Symposium*. Atlanta: TAPPI Press; 1997:453-458.
Pulper chemistry is demonstrated to impact process water clarification. Residual sodium silicate can reduce flocculant effectiveness decreasing ink separation effectiveness and increasing process water turbidity. The mechanism of this flocculant effectiveness reduction is thought to be reduction of the flocculant cationic charge density.
60. Sheridan GF, Rohlf EA, Aston D: **Preventing redeposition of ink in a deinking operation means improved recycled fibre.** *Proc TAPPI Pulping Conference*. Atlanta: TAPPI Press; 1992:487-492.
61. Röring A, Wackeberg E: **Characterization of deinking white water - influence on flotation and bleaching efficiency.** *Pulp Paper Can* 1997, 98:17-21.
62. Borchardt JK, Raney KH, Shpakoff PG, Matalamaki DW, Denley DR: **Insights into flexographic newsprint deinking: laboratory and pilot scale deinking studies.** *Proc TAPPI Pulping Conference*. Atlanta: TAPPI Press; 1994:1067-1103.
63. Upton BH, Krishnagopalan GA, Abubakr S: **Ultrafiltrative deinking of flexographic ONP: the role of surfactants.** *Proc TAPPI Recycling Symposium*. Atlanta: TAPPI Press; 1997:175-191.

Patents

- P1. Ishibashi Y, Myauchi Y, Inoe M, Edo T: **De-inking agents for the regeneration of wastepaper.** Japanese patent, 24/2/94; Kokai Tokkyo Koho 06 49 790.
- P2. Shiroishi T, Myauchi Y, Ishibashi Y, Takahashi H: **De-inking agents for the recycling of wastepaper.** Japanese patent, 2/2/93; Kokai Tokkyo Koho 05 25 782.
- P3. Ishibashi Y, Urushibata H: **Polyoxyalkylene compositions for de-inking wastepaper.** German patent, 12/3/92; 4 217 907.
- P4. Ishibashi Y, Urushibata H: **De-inking agents in paper regeneration.** Japanese patent, 10/14/92; Kokai Tokkyo Koho 04 289 287.
- P5. Okamoto Y, Hirakouchi Y, Hagiwara M: **Polyoxyalkylene composition for de-inking wastepaper by flotation method.** Canadian patent application, 3/1/93; 2 076 308.